with the transplant registry and will be on the waiting list.

When there is a heart or lung available for transplant and allotted for the patient on the waiting list, factors such as whether it matches the patient's blood group, weight and size are taken into consideration before the transplant.

PIONEERING MECHANICAL CIRCULATORY SUPPORT

While waiting for a suitable donor, some patients may be placed on LVAD (left ventricular assist device). These are primarily used to pump blood from the heart's left ventricle to the rest of the body while the patient is waiting for the heart transplant, but it can have long-term applications as well. "These pumps were initially introduced as a temporary bridge for a transplant and meant to last for a few months to around a year. Increasingly, they have become more sophisticated and can last for 10-15 years," says Dr KR Balakrishnan.

PAEDIATRIC HEART TRANSPLANTS: A SPECIAL CHALLENGE

MGM Healthcare runs the largest paediatric heart transplant programme in this part of the world. "We have done more than 140 paediatric heart transplants over the last decade including one of a six-month-old child," says Dr Balakrishnan.

Dr Balakrishnan gained international recognition for overseeing the first paediatric heart transplant surgery in India which was performed on a two-year nine-month-old international patient from Russia. This form of transplant comes with its own unique challenges. "Finding an organ for a child is not easy because, in India, most organ donations come from adults. We have to find a way to take a bigger heart from an older patient and fit it inside the chest of a small child," explains Dr Balakrishnan.

MAKING HISTORY DURING COVID-19

The COVID-19 pandemic was a challenging time for transplants, as organs weren't easily available, and there were no flights, leading to enormous logistical hurdles. In spite of this, Dr Balakrishnan and his team conducted a bilateral lung transplantation for a patient with COVID-19-associated ARDS (acute respiratory distress syndrome), one of the first 12 such procedures in the entire world. The medical team conducted the operation despite great risk to themselves, at a time when the COVID-19 vaccine had still not been developed. Subsequently, MGM Healthcare has conducted several such procedures.

Working in an integrated manner, MGM Healthcare's expert team of cardiologists, cardiac surgeons, cardiac anaesthesiologists, critical and intensive care physicians and rehab specialists provide the best care for patients with advanced heart failure, congenital heart disease, cardiomyopathy and other conditions.

POST-OPERATIVE CARE

In the case of transplants, post-operative care is as critical as the operation itself and immuno-suppressive drugs are used to prevent organ rejection. They have to be monitored carefully, treading a fine line. "If you suppress too much, you get infection; if you suppress too little, you get rejection. There's a science and an art to it," explains Dr Balakrishnan.

Patients need to exercise vigilance as well. "They need to understand that although they become functionally normal, it's not their own heart or lung inside their body, and they should respect it," says Dr Balakrishnan. Patients often feel so normal, there is a chance they can forget to take their medicines, especially when travelling on vacation or for a family function.

THE FUTURE OF TRANSPLANTS

As heart and lung transplants are not elective surgeries, they cannot be scheduled in advance and patients may have to wait several months for an organ. "The heart needs to be used within four hours of donation and geographical distances can sometimes pose a challenge," adds Dr Balakrishnan.

Thankfully, new technologies and machines are being developed which can keep a donated heart alive and viable for up 48 hours. MGM Healthcare, always at the forefront of medical technology, hopes to induct such machines in the next couple of years.

Working in an integrated manner, MGM Healthcare's expert team of cardiologists, cardiac surgeons, cardiac anaesthesiologists, critical and intensive care physicians and rehab specialists provide the best care for patients with advanced heart failure, congenital heart disease, cardiomyopathy and other conditions. This makes the Institute of Heart and Lung Transplant and Mechanical Circulatory Support one of the best choices in patients with complex heart and lung conditions.



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